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NUMERICAL METHODS

CPE 312

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Lecturer



RECALL

1. What is Numerical Methods?
2. What are the importance of Numerical Methods?
3. Give the differences between Numerical Methods and Analytical Methods.



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NONLINEAR, TRANSCENDENTAL, AND POLYNOMIAL FUNCTIONS TECHNIQUES



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Learning Objectives

- *Define Nonlinear, Transcendental and Polynomial Functions**
- *Indirect or Iterative Methods**
- *Bracketing Methods**
- *Graphical Method**



Nonlinear Equation – an equation is said to be nonlinear when it involves terms of degree higher than 1 in the unknown quantity.

Examples:

$$x^2 + xy = 10$$

$$y + 3xy^2 = 57$$



Transcendental Functions – is one that is non-algebraic. These include trigonometric, exponential, logarithmic, and other, less familiar functions.

Examples:

$$f(x) = \ln x^2 - 1$$

$$f(x) = e^{-x} - x$$



Polynomial Functions – are a simple class of algebraic functions that are represented generally by $f(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$ where n is the order of polynomials and a is the constant.

Examples:

$$f(x) = 1 - 7.23x + 2.5x^2$$

$$f(x) = 3x + 5x^2 - x^3 + 7x^6$$



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Indirect or Iterative Methods

An iterative method is a mathematical procedure that uses an initial value to generate a sequence of improving approximate solutions for a class of problems.



- **Bracketing Method** – deals with methods that exploit the fact that a function typically changes sign in the vicinity of a root. These techniques are called “bracketing” because two initial guesses for the root are required.



1. Graphical Method

A simple method for obtaining an estimate of the root of the equation $f(x) = 0$ is to make a plot of a function and observe where it crosses the x-axis.



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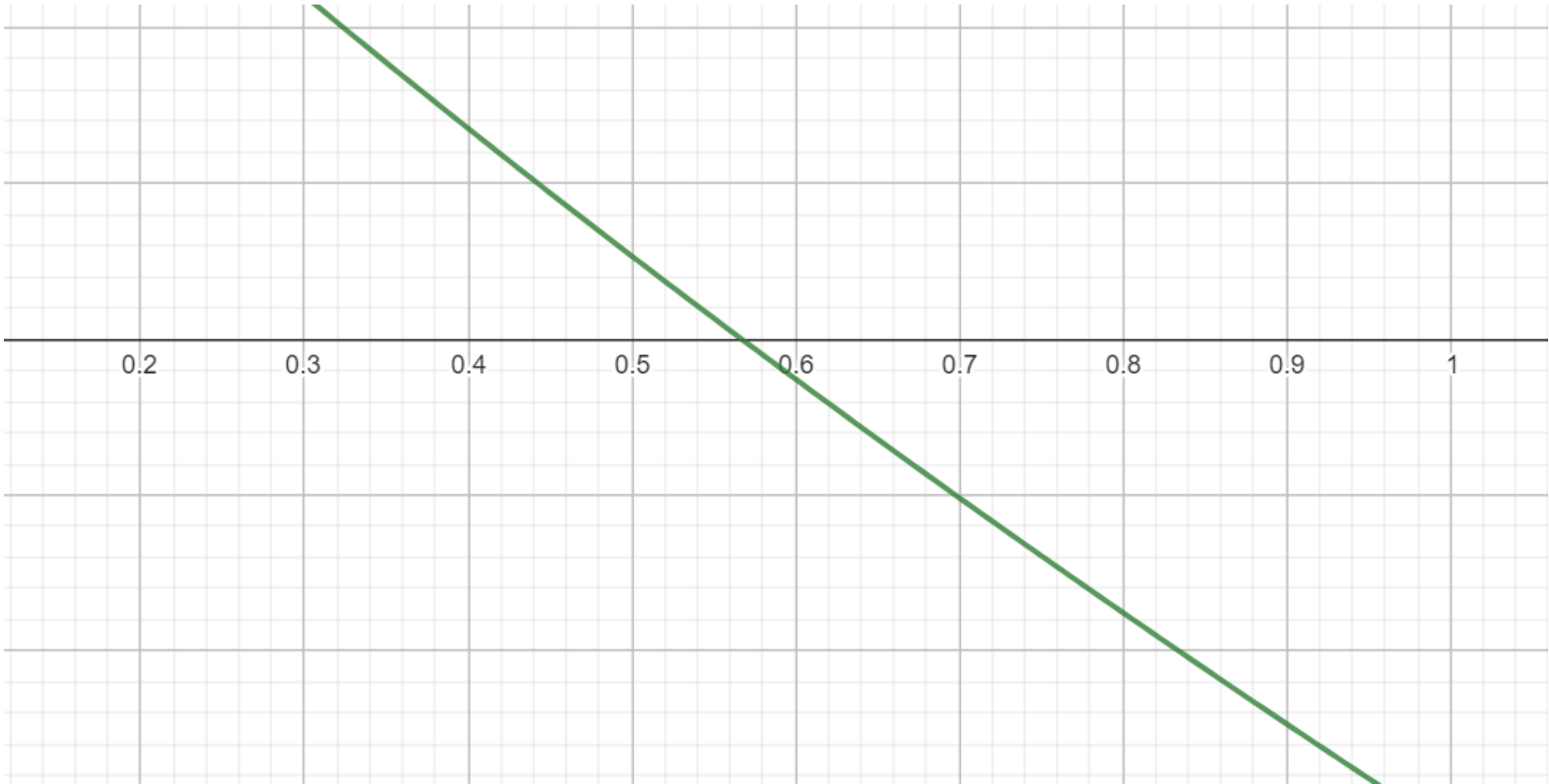
Illustrative Example #1:

Use the graphical approach to determine the root of the equation
$$f(x) = e^{-x} - x.$$



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x	$f(x) = e^{-x} - x$
0	
0.1	
0.2	
0.3	
0.4	
0.5	
0.6	
0.7	



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x	$f(x) = e^{-x} - x$
0	1
0.1	
0.2	
0.3	
0.4	
0.5	
0.6	
0.7	



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x	$f(x) = e^{-x} - x$
0	1
0.1	0.80484
0.2	
0.3	
0.4	
0.5	
0.6	
0.7	



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x	$f(x) = e^{-x} - x$
0	1
0.1	0.80484
0.2	0.61873
0.3	
0.4	
0.5	
0.6	
0.7	



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x	$f(x) = e^{-x} - x$
0	1
0.1	0.80484
0.2	0.61873
0.3	0.44082
0.4	
0.5	
0.6	
0.7	



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x	$f(x) = e^{-x} - x$
0	1
0.1	0.80484
0.2	0.61873
0.3	0.44082
0.4	0.27032
0.5	
0.6	
0.7	



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x	$f(x) = e^{-x} - x$
0	1
0.1	0.80484
0.2	0.61873
0.3	0.44082
0.4	0.27032
0.5	0.10653
0.6	
0.7	



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x	$f(x) = e^{-x} - x$
0	1
0.1	0.80484
0.2	0.61873
0.3	0.44082
0.4	0.27032
0.5	0.10653
0.6	-0.05119
0.7	



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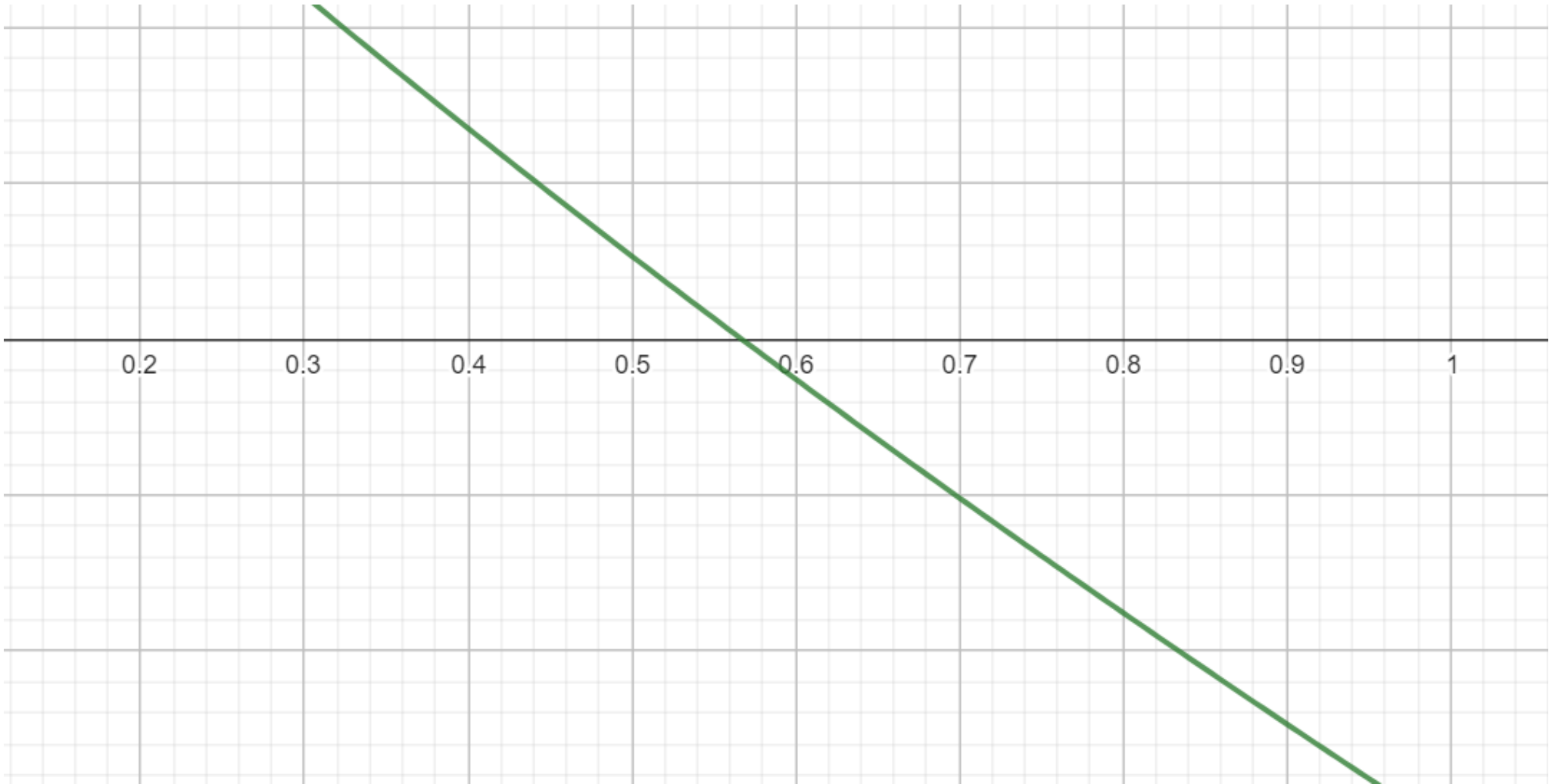
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x	$f(x) = e^{-x} - x$
0	1
0.1	0.80484
0.2	0.61873
0.3	0.44082
0.4	0.27032
0.5	0.10653
0.6	-0.05119
0.7	-0.20341



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$$x = 0.56$$

$$f(x) = e^{-x} - x$$

$$f(0.56) = e^{-0.56} - 0.56$$



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$$x = 0.56$$

$$f(x) = e^{-x} - x$$

$$f(0.56) = e^{-0.56} - 0.56$$

$$f(0.56) = 0.01121$$



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$$x = 0.56$$

$$f(x) = e^{-x} - x$$

$$f(0.56) = e^{-0.56} - 0.56$$

$$f(0.56) = 0.01121$$

$$x = 0.57$$

$$f(x) = e^{-x} - x$$

$$f(0.57) = e^{-0.57} - 0.57$$



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$$x = 0.56$$

$$f(x) = e^{-x} - x$$

$$f(0.56) = e^{-0.56} - 0.56$$

$$f(0.56) = 0.01121$$

$$x = 0.57$$

$$f(x) = e^{-x} - x$$

$$f(0.57) = e^{-0.57} - 0.57$$

$$f(0.57) = -0.00447$$



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$$x = 0.56$$

$$f(x) = e^{-x} - x$$

$$f(0.56) = e^{-0.56} - 0.56$$

$$f(0.56) = 0.01121$$

$$x = 0.57$$

$$f(x) = e^{-x} - x$$

$$f(0.57) = e^{-0.57} - 0.57$$

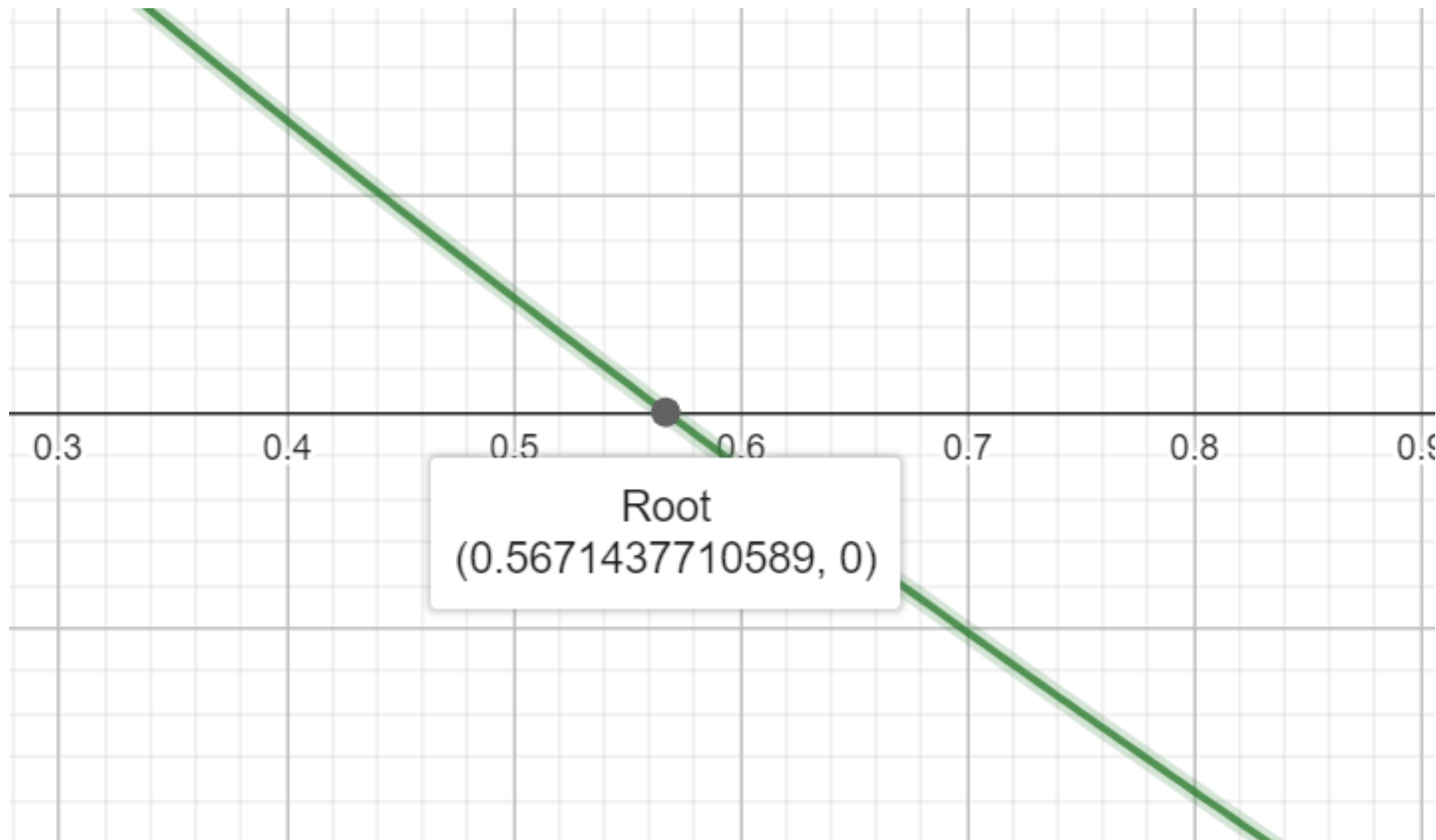
$$f(0.57) = -0.00447$$

The approximate root of $f(x) = e^{-x} - x$ is 0.57.



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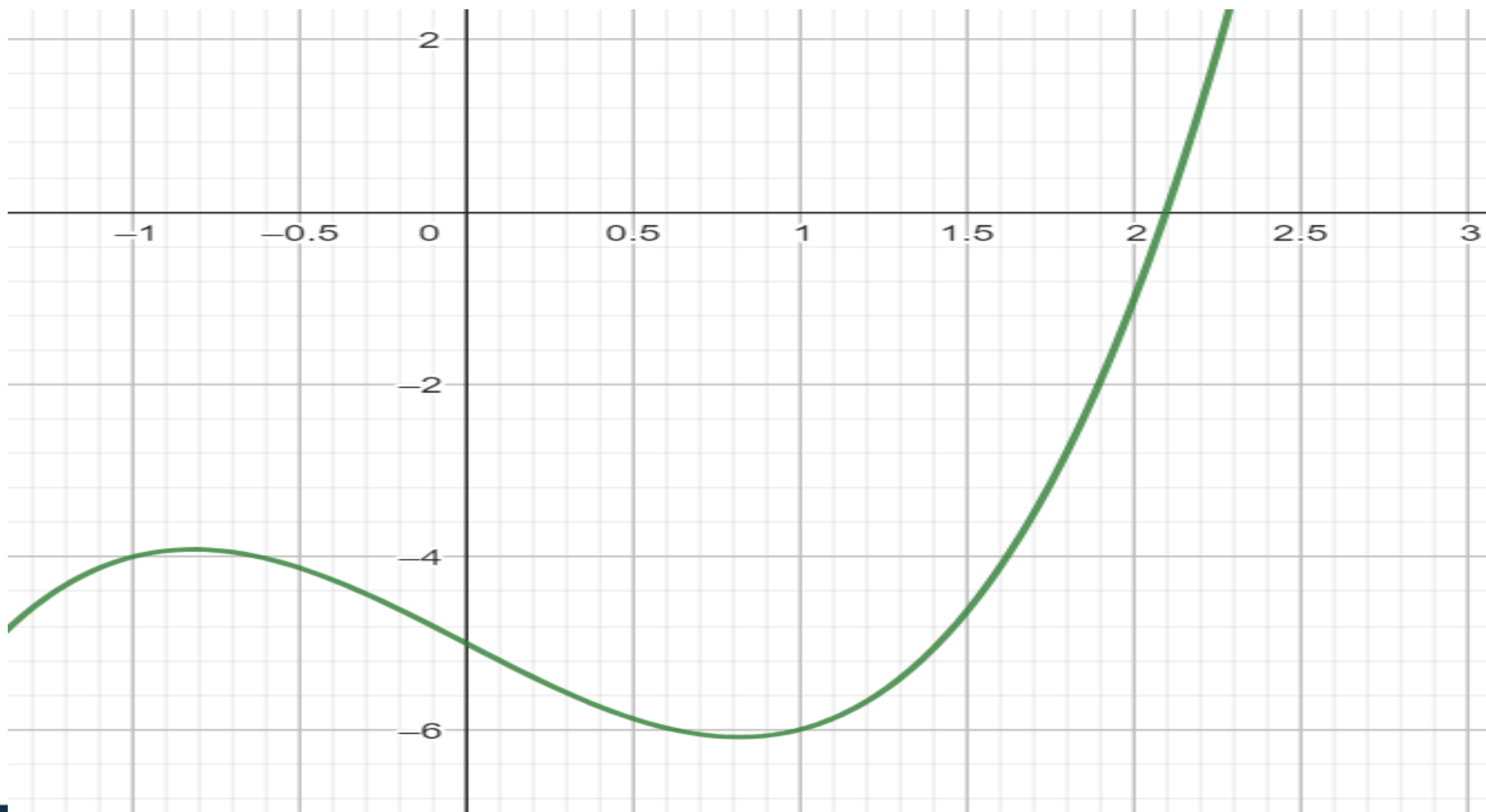
Illustrative Example #2:

Use the graphical approach to determine the root of the equation
$$f(x) = x^3 - 2x - 5.$$



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[illegible]

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x	$f(x) = x^3 - 2x - 5$
2	-1
2.03	-0.69457
2.07	-0.27026
2.08	



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x	$f(x) = x^3 - 2x - 5$
2	-1
2.03	-0.69457
2.07	-0.27026
2.08	-0.16109



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x	$f(x) = x^3 - 2x - 5$
2	-1
2.03	-0.69457
2.07	-0.27026
2.08	-0.16109
2.09	



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x	$f(x) = x^3 - 2x - 5$
2	-1
2.03	-0.69457
2.07	-0.27026
2.08	-0.16109
2.09	-0.05067



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x	$f(x) = x^3 - 2x - 5$
2	-1
2.03	-0.69457
2.07	-0.27026
2.08	-0.16109
2.09	-0.05067
2.1	



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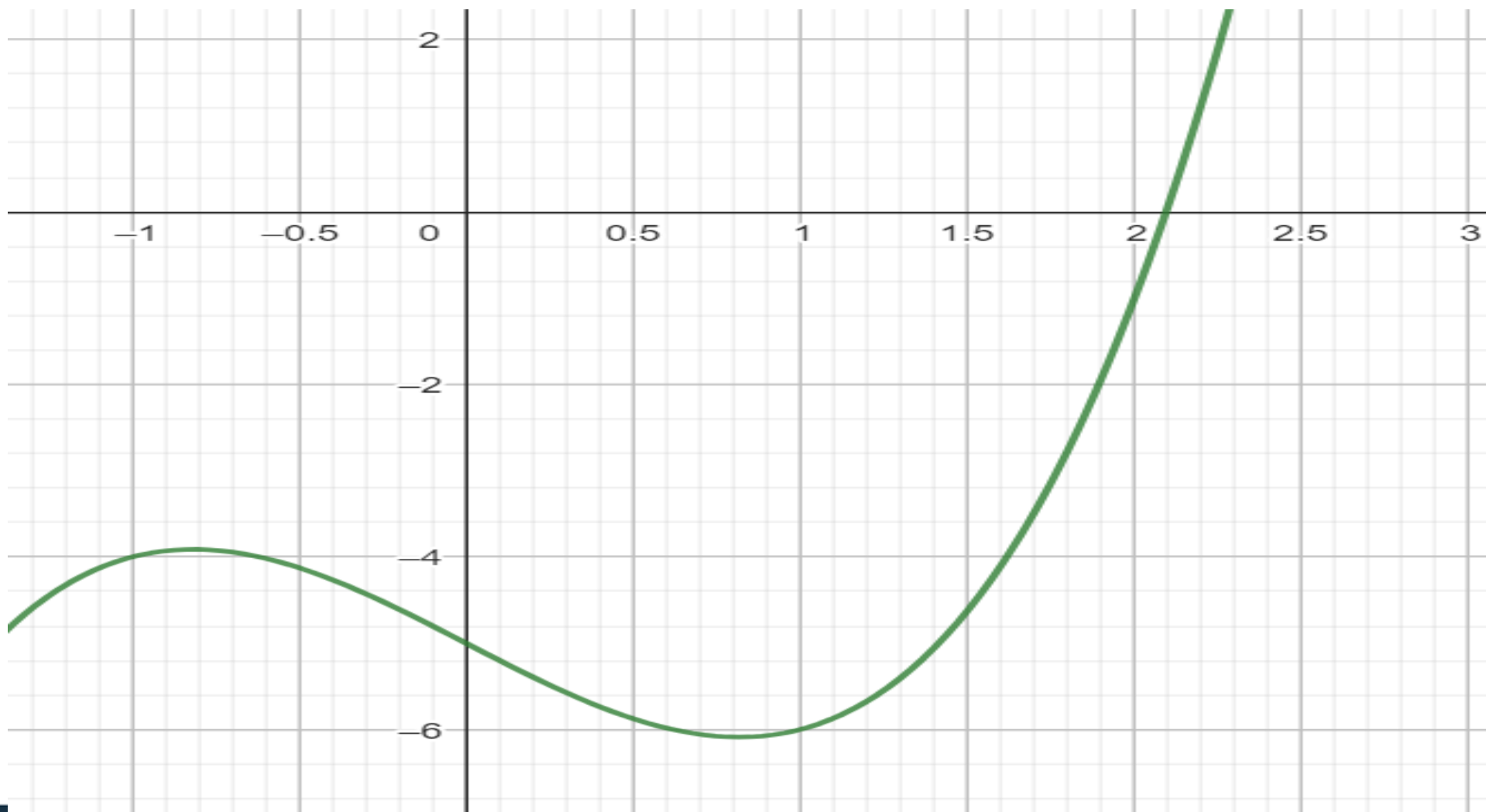
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x	$f(x) = x^3 - 2x - 5$
2	-1
2.03	-0.69457
2.07	-0.27026
2.08	-0.16109
2.09	-0.05067
2.1	0.061



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$$x = 2.094$$

$$f(x) = x^3 - 2x - 5$$

$$f(2.094) = (2.094)^3 - 2(2.094) - 5$$



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$$x = 2.094$$

$$f(x) = x^3 - 2x - 5$$

$$f(2.094) = (2.094)^3 - 2(2.094) - 5$$

$$\mathbf{f(2.094) = -0.00615}$$



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$$x = 2.095$$

$$f(x) = x^3 - 2x - 5$$

$$f(2.095) = (2.095)^3 - 2(2.095) - 5$$



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$$x = 2.095$$

$$f(x) = x^3 - 2x - 5$$

$$f(2.095) = (2.095)^3 - 2(2.095) - 5$$

$$f(2.095) = 0.00501$$



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$$x = 2.095$$

$$f(x) = x^3 - 2x - 5$$

$$f(2.095) = (2.095)^3 - 2(2.095) - 5$$

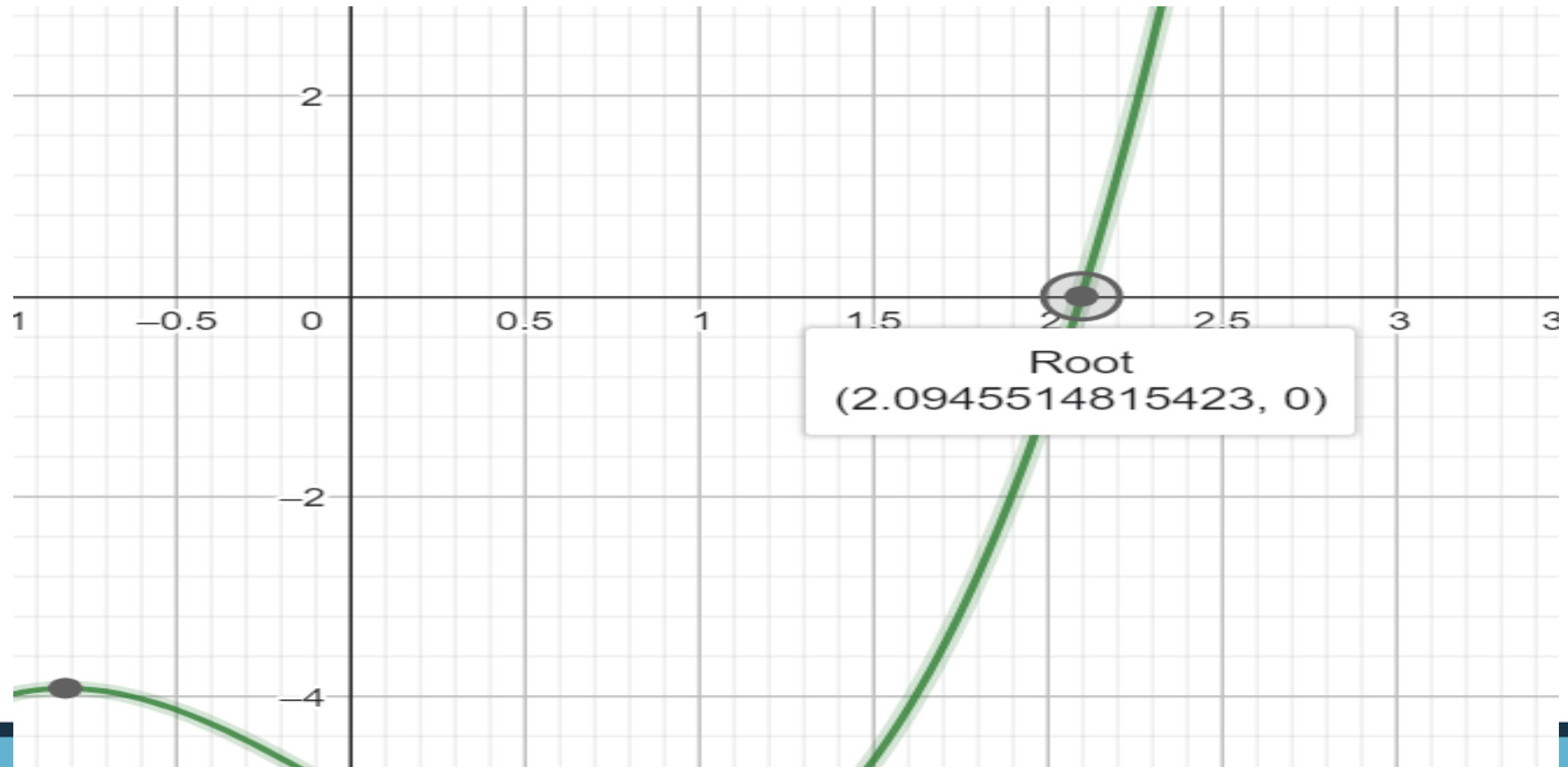
$$f(2.095) = 0.00501$$

The approximate root of $f(x) = x^3 - 2x - 5$ is 2.095.



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ACTIVITY # 3



Use Graphical Method to approximate the roots of the following functions. (*Note: Construct a table of values for x and $f(x)$, sketch the graph of $f(x)$ showing the root/s, and check the approximated solution which is closest to $f(x) = 0$.*)



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1. $f(x) = 3x + 5x^2 - x^3 + 7x^6$
2. $f(x) = \ln x^2 - 1$



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